

Patents & Trademarks

The intellectual property of Derek F. Amanatullah, M.D., Ph.D. spans operative safety and efficiency, mixed-reality implant placement, revision hip and knee fixation, soft-tissue tensioning, bacteriophage delivery, and dormant-infection detection. The portfolio includes issued patents, pending applications, and trademarks held by Stanford University and five founded companies.

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ISSUED PATENTS

32

FILINGS & MARKS

5

COMPANIES

1

EXCLUSIVE LICENSE

Assigned to Stanford University 6

- System and Method for Detecting, Characterizing, and Preventing Unintentional Tissue Damage During a Surgical Procedure**
#62/991,546, #17/206,012 (allowed), PCT-US21/23028
- Method for Monitoring Object Flow within a Surgical Space During a Surgery**
#62/875,283, #16/533,632, #16/932,607, #17/962,316, #18/085,509, #11,583,360, #11,666,409, PCT-US20/42661
- Controlled Release of Bacteriophage to Treat Implant Infections**
#63/024,963, #18/552,406, #EU4314120, #PCT-US2022/071351
- Methods for Detecting a Dormant Infection**
#63/677,070, PCT-US25/039107
- Method for Detecting Subclinical Bacterial Infections**
#63/848,642
- Method and Device for Detecting Dormant and Active Infection**
#63/6965,953

Assigned to Arthrology Consulting, LLC 7

- Design of Repetitive-A Jewelry Piece**
#29/594,948, #D873701
- Cool-cut Cast Saw Blade**
#14/102,403, #9,468,570
- Modular Total Knee Arthroplasty System and Method**
#62/318,684, #15/477,578, #16/209,888, #16/833,194, #17/661,629, #18/908,005, #10,182,830, #10,639,047, #11,344,320, #12,114,871
- Expandable Augment System for Acetabular Cup**
#17/455,893, #18/590,344, #19/229,431, #11,944,544, #12,343,259
- Off-center Liner for Acetabular Cup**

#17/455,130, #18/511,283, #11,857,423

Distractor Tool and System for Balancing a Hip Joint in Hip Joint Replacement Surgery

#63/896,174

Phase-shifted Femoral Heads

#63/896,154

Under Exclusive License to Medacta 4

Method for Augmenting a Surgical Field with Virtual Guidance Content

#62/328,330, #62/612,895, #15/499,046, #15/594,623, #16/588,926, #10,194,990, PCT-US18/29506

Method for Augmenting a Surgical Field with Virtual Guidance and Tracking and Adapting to Deviations from a Surgical Plan

#16/238,500, #16/238,504, #17/575,545, #10,806,518, #10,813,700, #11,253,321, PCT-US19/12111

Method for Detecting and Tracking Deviations from a Surgical Plan

#62/612,901

Method for Patient Registration and Planning for Surgical Customization, Visualization, and Evaluation, of Intra Operative Deviation and Post Execution Accuracy

#62/363,022

Assigned to Arthrology Designs, LLC (DBA: PlantarTech) 4

Trademark PlantarTech (Class 10: Splints for Feet, #7,249,638)

Trademark Logo (Class 10: Splints for Feet, #6,687,181)

Dynamic Hallux Tension Device for Plantar Fasciitis

#13/745768, #15/686,515, #9,320,637, #9,750,631, #10,758,395

Foot Splint

#29/788,488, #D1082015

Assigned to Knimble Designs, Inc. 5

Methods for Generating Intraoperative Surgical Guidance during Joint Arthroplasty Based on Dynamic Ligament Tension

#62/853,600, #16/886,413, #17/947,670, #18/677,753, #11,478,302, #12,023,104

System for Total Knee Replacement

#63/108,243, #17/515,205, #18/419,350, #11,911,050

Method and Apparatus for Mid-flexion Balancing during Knee Arthroplasty

#18/127,250, #18/242,612, #18/503,600, #11,786,262, #11,849,960, PCT-US24/20892, WO-2024/206057, AUS-2024/254804, CAN-3,286,912, JAP-2025/556580, EPO-24781587.1

System for Identifying Positions of an Artificial Tibial Component during Total Knee Arthroplasty

#63/523,303, #18/755,524, #12,648,861

Method and Apparatuses for Performing a Knee Arthroplasty

#63/575,092, #63/672,332, #19/170,618, #19/263,224 (allowed), #19/263,650 (allowed), #19/264,495, PCT-US25/23209

Assigned to Zooly Labs, Inc. 2

System for Balancing a Hip During a Hip Arthroplasty

#63/321,013, #18/123,177, PCT-US23/15552

Combined Sensing and Adjustment Tool for Hip Arthroplasty Balancing

#63/693,135, #19/325,361

Assigned to nSight Surgical, Inc. 4

Surgical Imaging System and Method

#63/285,466, #63/285,467, #18/074,408

Multicamera Tracking Set-up: Visual Tracking of Lap Sponges, Needles Across OR

#63/347,144

Method for Tracking Objects within a Surgical Space

#63/429,509

Method for Predicting Consumption of Surgical Consumables During Surgical Operation

#63/529,117, #18/786,053